

Transient Transport Equations for Energy Propagation in Visco-Elastic Structures

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Abstract

We apply some rigorous mathematical results obtained in the frame of microlocal analysis to derive the exact propagation features of the high-frequency energy of randomly heterogeneous elastic structures. The theory shows that the energy observables satisfy transport equations, instead of the commonly assumed diffusion equations invoked by the structural-acoustics community. Such a diffusion limit can be obtained from transport phenomena only under certain circumstances which are outlined in the paper. Our main focus is to extend this theory to the case of visco-elastic materials with memory effect: the transport equations remain basically unchanged in this case. We consider the example of a thick circular cylindrical shell to illustrate it.

1. Introduction

Complex industrial structures exhibit typical transport and diffusive behaviours in the high-frequency range of vibration [1]. Their dynamics are physically well represented by global standing waves (the eigenmodes) in the low-frequency range, but this approach fails dramatically as frequency increases because of the multiple interactions of waves with material heterogeneities at the wavelength scale, and reflections on the boundaries and at the interfaces between structural components with different stiffnesses (stiffeners on plates and shells, bulkheads, stringers ...). Predictive methods for high-frequency vibrations are needed in the automotive and aerospace industries for applications to noise control, acoustic comfort and discretion, or pyrotechnic shock simulation for instance. Statistical Energy Analysis (SEA, see [2]) is the most popular method among engineers to deal with such problems. It is based on some rigorous arguments developed in the late 50s, but also on several phenomenological assumptions which can hardly be satisfied exactly and the theory itself is essentially an approximation. Vibrational conductivity analogy for high-frequency structural dynamics has attracted several researchers in the early 90s [3], but this model is based on unproven arguments as well (the theory itself is in fact essentially incorrect).

In [4], we have used the radiative transfer model proposed in [5] to analyze the high-frequency vibrations of randomly heterogeneous plates. This model is based on the assumptions that (i) wavelengths are short with respect to the macroscopic features of the medium (the high-frequency range), (ii) they are comparable to the correlation lengths of random heterogeneities of the medium, and (iii) fluctuations of the latter are small and rather isotropic. In this paper we focus on the extension of the theory to the case of visco-elastic materials with memory effect and its application to shells. We also outline how generalized diffusion equations can be derived from the transfer model. These are the correct equations that should be considered in the vibrational conductivity analogy, and we give the minimum hypotheses for the latter to be applicable.

In the next section we recall the basic results of the transport theory of energy for visco-elastic media. The influence of random heterogeneities is described in section 3, leading to the diffusive regime outlined in section 4. Section 5 shows how shell equations can be put in the proposed framework.

2. Wigner measures as energy density estimates

We summarize in this section the basic developments of Ryzhik *et al* [5], Gérard *et al.* [6] and Akian [7]. The interested reader shall consult these references for a detailed presentation of the theory and, of course, all proofs of the results given below. Consider a first-order hyperbolic system of the form

$$\mathbf{A}(\mathbf{x}, 0) \partial_t \mathbf{v} + \sum_{j=1}^d \mathbf{D}_j \partial_{x_j} (\mathbf{v} + \mathbf{B}(\mathbf{x}, \cdot) * \mathbf{v}) = \mathbf{0} \quad (1)$$

where $\mathbf{v}$ is the vector of state variables in $\mathbb{C}^n$, and $\mathbf{x} \in \mathbb{R}^d$ runs through the physical space; $\mathbf{A}(\mathbf{x}, t)$ is a symmetric, positive definite $n \times n$ real matrix; $\mathbf{D}_j$ are symmetric $n \times n$ real matrices independent of position $\mathbf{x}$ and time t ; $\mathbf{B}(\mathbf{x}, t)$ is a symmetric $n \times n$ real matrix; and $*$ is the time convolution product $f * g(t) = \int_0^t f(\tau)g(t - \tau)d\tau$,

$t \geq 0$. We denote by $\mathbf{v}^\epsilon$ the solution of eq. (1) for initial conditions oscillating rapidly (in the sense precised below) at an arbitrarily small scale ϵ , considering slowly varying mechanical coefficients such that neither $\mathbf{A}$ nor $\mathbf{B}$ depend on ϵ . We also assume that the sequence $(\mathbf{v}^\epsilon)$ is uniformly bounded in $(L^2(\mathbb{R}^d))^n$, where $L^2(\mathbb{R}^d)$ is the set of square integrable functions with respect to the space variable.

The setting of eq. (1) is very general for wave equations arising from physical problems: it embeds Maxwell's equations for electromagnetic waves, the wave equation for acoustics or the Navier equation for three-dimensional elastodynamics. For an isotropic visco-elastic medium characterized by its time-dependent Lamé's moduli $\lambda(\mathbf{x}, t)$ and $\mu(\mathbf{x}, t)$ and density $\rho(\mathbf{x})$, the Navier equation of motion for the displacement vector $\mathbf{u} \in \mathbb{R}^3$ is:

$$\rho \partial_t^2 \mathbf{u} = \text{Div} \boldsymbol{\sigma}(\mathbf{u})$$

where the stress tensor $\boldsymbol{\sigma}$ is linked to the strain tensor $\boldsymbol{\epsilon} = \nabla \otimes_s \mathbf{u}$ by:

$$\begin{aligned} \boldsymbol{\sigma} &= \lambda(\mathbf{x}, 0) \text{div} \mathbf{u} \mathbf{I}_3 + 2\mu(\mathbf{x}, 0) \boldsymbol{\epsilon} \\ &+ \partial_t \lambda(\mathbf{x}, \cdot) * \text{div} \mathbf{u} \mathbf{I}_3 + 2\partial_t \mu(\mathbf{x}, \cdot) * \boldsymbol{\epsilon}. \end{aligned}$$

Here $\mathbf{I}_j$ stands for the $j \times j$ identity matrix. These equations have the form of eq. (1) setting $\mathbf{v} = (\partial_t u_1, \partial_t u_2, \partial_t u_3, \epsilon_{11}, \epsilon_{22}, \epsilon_{33}, \epsilon_{23}, \epsilon_{13}, \epsilon_{12}, p)^T$, with $\epsilon_{ij} = \mu(\mathbf{x}, 0)(\partial_{x_j} u_i + \partial_{x_i} u_j)$ and $p = \lambda(\mathbf{x}, 0) \text{div} \mathbf{u}$, together with $\mathbf{A}(\mathbf{x}, t) = \text{diag}(\rho \mathbf{I}_3, 1/2\mu \mathbf{I}_3, 1/\mu \mathbf{I}_3, 1/\lambda)$ and $\mathbf{B}(\mathbf{x}, t) = \partial_t \mathbf{A}^{-1}(\mathbf{x}, t) \mathbf{A}(\mathbf{x}, 0)$. As shown in [7], the results below extend to this case without any modification induced by memory effects assuming that the material is synchronous as well (Poisson's coefficient remains independent of t).

We inject high-frequency energy into the system described by eq. (1) through, say, initial conditions of the form

$$\mathbf{v}^\epsilon(\mathbf{x}, 0) = \mathbf{v}_0^\epsilon(\mathbf{x}), \quad (2)$$

where the data $\mathbf{v}_0^\epsilon$ are bounded in $(L^2(\mathbb{R}^d))^n$ as ϵ goes to 0 and oscillating at most at the frequency $\frac{1}{\epsilon}$ (the general conditions to be fulfilled by the initial data for the following to apply are given in [6]). The mechanical energy density associated to the solution $\mathbf{v}^\epsilon$ of eqs. (1)–(2) is defined by

$$\mathcal{E}^\epsilon(\mathbf{x}, t) = \frac{1}{2} (\mathbf{v}^\epsilon, \mathbf{v}^\epsilon)_{\mathbf{A}(\mathbf{x}, 0)} \quad (3)$$

while the densities of power flow along each direction $j \in \{1, 2, \dots, d\}$ are

$$\Pi_j^\epsilon(\mathbf{x}, t) = \frac{1}{2} (\mathbf{v}^\epsilon, \mathbf{v}^\epsilon)_{\mathbf{D}_j} \quad (4)$$

with the notation $(\mathbf{u}, \mathbf{v})_{\mathbf{M}} = (\mathbf{M}\mathbf{u}, \mathbf{v})$. A potential good candidate to study the high-frequency energetic features

of the system (1)–(2) is the Wigner transform, because it has a direct link with the above quantities in the limit $\epsilon \rightarrow 0$, as well as some other nice evolution properties given below. The $n \times n$ matrix-valued Wigner transform $\mathbf{W}^\epsilon[\mathbf{v}]$ of a function $\mathbf{v}$ lying in a bounded subset of $(L^2(\mathbb{R}^d))^n$ is defined by:

$$\begin{aligned} \mathbf{W}^\epsilon[\mathbf{v}](\mathbf{x}, \mathbf{k}, t) &= \\ \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} e^{i\mathbf{k} \cdot \mathbf{y}} \mathbf{v} \left(\mathbf{x} - \frac{1}{2} \epsilon \mathbf{y}, t \right) \mathbf{v}^* \left(\mathbf{x} + \frac{1}{2} \epsilon \mathbf{y}, t \right) d\mathbf{y} \end{aligned} \quad (5)$$

where $\mathbf{v}^*$ stands for the conjugate transpose. Then $\mathbf{W}^\epsilon[\mathbf{v}^\epsilon]$ for $\mathbf{v}^\epsilon$ of (1)–(2) given above has an hermitian weak limit (in the sense of tempered distributions) $\mathbf{W}^\epsilon[\mathbf{v}^\epsilon] \rightharpoonup \mathbf{W}$ as $\epsilon \rightarrow 0$ up to an extracted subsequence which is also a non-negative measure [6], the so-called Wigner measure. We assume that $\mathbf{v}^\epsilon$ has the same oscillatory properties as the initial conditions, then owing to Proposition 1.7 of [6] it can be shown that the energy $\mathcal{E}(\mathbf{x}, t)$ and power flow $\Pi_j(\mathbf{x}, t)$ densities in the limit $\epsilon \rightarrow 0$ are estimated from its Wigner measure by:

$$\mathcal{E}(\mathbf{x}, t) = \frac{1}{2} \int_{\mathbb{R}^d} \mathbf{A}(\mathbf{x}, 0) : \mathbf{W}(\mathbf{x}, \mathbf{k}, t) d\mathbf{k} \quad (6)$$

and for $j \in \{1, 2, \dots, d\}$:

$$\Pi_j(\mathbf{x}, t) = \frac{1}{2} \int_{\mathbb{R}^d} \mathbf{D}_j : \mathbf{W}(\mathbf{x}, \mathbf{k}, t) d\mathbf{k}, \quad (7)$$

respectively, where $\mathbf{M} : \mathbf{N} = \text{Tr}(\mathbf{M}\mathbf{N}^T)$ stands for the usual tensor scalar product and Tr is the trace of a matrix.

We now turn to the evolution properties of the Wigner measure. Let:

$$\mathbf{L}(\mathbf{x}, \mathbf{k}) = \mathbf{A}^{-1}(\mathbf{x}, 0) \sum_{j=1}^d k_j \mathbf{D}_j. \quad (8)$$

This is the dispersion matrix of the above system, which is self-adjoint with respect to the scalar product $(\mathbf{u}, \mathbf{v})_{\mathbf{A}(\mathbf{x}, 0)}$. Thus its n right eigenvectors form an orthonormal family with respect to this scalar product. The latter and the associated real eigenvalues are denoted by $\mathbf{b}_{\alpha_r}(\mathbf{x}, \mathbf{k})$ and $\lambda_{\alpha_r}(\mathbf{x}, \mathbf{k})$, respectively, for $1 \leq \alpha \leq M \leq n$ and $1 \leq r \leq R_\alpha$, where R_α is the order of multiplicity of eigenvalue λ_α for a mode α among the M existing ones, such that $\sum_{\alpha=1}^M R_\alpha = n$. We assume that R_α does not depend on $(\mathbf{x}, \mathbf{k})$. Then it can be shown that the Wigner measure writes:

$$\mathbf{W}(\mathbf{x}, \mathbf{k}, t) = \sum_{\alpha=1}^M \sum_{r,s=1}^{R_\alpha} w_{\alpha}^{rs}(\mathbf{x}, \mathbf{k}, t) \mathbf{b}_{\alpha_r}(\mathbf{x}, \mathbf{k}) \mathbf{b}_{\alpha_s}^*(\mathbf{x}, \mathbf{k})$$

and if we define coherence matrices $\mathbf{W}_\alpha$, $1 \leq \alpha \leq M$, by:

$$W_{\alpha,rs}(\mathbf{x}, \mathbf{k}, t) = w_{\alpha}^{rs}(\mathbf{x}, \mathbf{k}, t), \quad 1 \leq r, s \leq R_\alpha,$$

then they satisfy the following transport equations:

$$\partial_t \mathbf{W}_\alpha + \{\lambda_\alpha, \mathbf{W}_\alpha\} + [\mathbf{W}_\alpha, \mathbf{N}_\alpha] = \mathbf{0}. \quad (9)$$

$\{f, g\} = \nabla_{\mathbf{k}} f \cdot \nabla_{\mathbf{x}} g - \nabla_{\mathbf{x}} f \cdot \nabla_{\mathbf{k}} g$ stands for the usual Poisson's bracket and we define $[\mathbf{M}, \mathbf{N}] = \mathbf{M}\mathbf{N} + \mathbf{N}^* \mathbf{M}^*$. The $\mathbf{N}_\alpha$'s are skew-symmetric rotation matrices arising from the curvature of rays in heterogeneous media. They are identically zero for homogeneous media, or for the shell equations considered in section 5.

3. Influence of random heterogeneities

In this section we consider some randomness of the mechanical properties of the medium and follow again the developments of Ryzhik *et al* [5]. In their work these authors introduce a random perturbation $(\mathbf{V}(\mathbf{x}), \mathbf{x} \in \mathbb{R}^d)$ of matrix $\mathbf{A}(\mathbf{x}, 0)$ of the form:

$$\mathbf{A}(\mathbf{x}, 0) = \underline{\mathbf{A}}(\mathbf{x}, 0) \left[\mathbf{I}_n + \sqrt{\epsilon} \mathbf{V} \left(\frac{\mathbf{x}}{\epsilon} \right) \right]. \quad (10)$$

The magnitude $\sqrt{\epsilon}$ of perturbations is chosen such that they have comparable effects to the unperturbed background medium characterized by the mean matrix of parameters $\underline{\mathbf{A}}(\mathbf{x}, 0)$. However they need not be too strong, for in this case localization rather than transport may occur. Their correlation lengths are chosen to be comparable to wavelengths ϵ in order to allow strong interactions between the waves and the heterogeneities. In order that the system (1) remains symmetric hyperbolic the perturbation matrix has to satisfy the condition:

$$\underline{\mathbf{A}}(\mathbf{x}, 0) \mathbf{V}(\mathbf{y}) = \mathbf{V}^*(\mathbf{y}) \underline{\mathbf{A}}(\mathbf{x}, 0)$$

for all $\mathbf{x}$ and $\mathbf{y}$ of $\mathbb{R}^d$. Parameter fluctuations are modeled by a statistically homogeneous, second order matrix-valued random process with mean zero whose covariance functions are denoted by $R_{jklm}(\mathbf{x})$ and such that:

$$\begin{aligned} R_{jklm}(\mathbf{x} - \mathbf{y}) &= \mathbb{E} \{ V_{jk}(\mathbf{x}) V_{lm}(\mathbf{y}) \} \\ &= \int_{\mathbb{R}^d} e^{-i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})} S_{jklm}(\mathbf{k}) d\mathbf{k} \end{aligned}$$

where $\mathbb{E}\{\cdot\}$ stands for mathematical expectation. $\mathbf{k} \mapsto S_{jklm}(\mathbf{k})$ are the spectral density functions of random matrix $(\mathbf{V}(\mathbf{x}), \mathbf{x} \in \mathbb{R}^d)$ for $j, k, l, m \in \{1, 2, \dots, n\}$ and let $\mathbf{S}$ be the fourth-order spectral density tensor corresponding to these functions. Then it can be shown that the coherence matrices $\mathbf{W}_\alpha$, $1 \leq \alpha \leq M$, satisfy coupled radiative transfer equations given by:

$$\begin{aligned} \partial_t \mathbf{W}_\alpha + \{\lambda_\alpha, \mathbf{W}_\alpha\} + [\mathbf{W}_\alpha, \mathbf{N}_\alpha] &= \\ \sum_{\beta=1}^M \int_{\mathbb{R}^d} [\sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}') \mathbf{W}_\beta(\mathbf{k}')] \delta(\lambda_\alpha(\mathbf{k}) - \lambda_\beta(\mathbf{k}')) d\mathbf{k}' & \\ - [\Sigma_\alpha, \mathbf{W}_\alpha] & \quad (11) \end{aligned}$$

The differential scattering operators $\sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}')$ are defined by:

$$[\sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}') \mathbf{W}_\beta(\mathbf{k}')] = 2\pi \lambda_\alpha^2(\mathbf{k}) \times \mathcal{S}_{\alpha\beta}(\mathbf{k}, \mathbf{k}') \mathbf{W}_\beta(\mathbf{k}') \quad (12)$$

where $\mathcal{S}_{\alpha\beta}(\mathbf{k}, \mathbf{k}')$ is a fourth-order tensor with elements

$$\begin{aligned} \mathcal{S}_{rstu}^{\alpha\beta}(\mathbf{k}, \mathbf{k}') &= \\ \mathbf{c}_{\alpha_r}(\mathbf{k}) \mathbf{b}_{\beta_t}^*(\mathbf{k}') : \mathbf{S}(\mathbf{k} - \mathbf{k}') : \mathbf{c}_{\alpha_s}(\mathbf{k}) \mathbf{b}_{\beta_u}^*(\mathbf{k}'). \end{aligned}$$

The total scattering operators $\Sigma_\alpha(\mathbf{k})$ are defined by:

$$\begin{aligned} \Sigma_\alpha(\mathbf{k}) &= \\ \frac{1}{2} \sum_{\beta=1}^M \left\{ \int_{\mathbb{R}^d} [\sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}') \mathbf{I}_{R_\beta}] \delta(\lambda_\alpha(\mathbf{k}) - \lambda_\beta(\mathbf{k}')) d\mathbf{k}' \right. & \\ \left. - i \int_{\mathbb{R}^d} \frac{1}{\lambda_\alpha(\mathbf{k}) - \lambda_\beta(\mathbf{k}')} [\sigma_{\alpha\beta}(\mathbf{k}, \mathbf{k}') \mathbf{I}_{R_\beta}] d\mathbf{k}' \right\}. \quad (13) \end{aligned}$$

$\mathbf{b}_{\alpha_r}$ are the right eigenvectors of the mean dispersion matrix and $\mathbf{c}_{\alpha_r}$ are its left eigenvectors, given by $\mathbf{c}_{\alpha_r} = \underline{\mathbf{A}} \mathbf{b}_{\alpha_r}$, for $1 \leq \alpha \leq M$ and $1 \leq r \leq R_\alpha$. The scattering cross-section $\sigma_{\alpha\beta}(\mathbf{x}, \mathbf{k}, \mathbf{k}')$ gives the rate at which the energy mode β with wave vector $\mathbf{k}'$ is converted to the energy mode α with wave vector $\mathbf{k}$, at position $\mathbf{x}$; the total scattering cross-section $\Sigma_\alpha(\mathbf{x}, \mathbf{k})$ gives the rate at which the energy mode α with wave vector $\mathbf{k}$ at position $\mathbf{x}$ is converted to all other modes or wave numbers.

4. Diffusive regime

Transported waves exhibit a diffusive behaviour inasmuch as their direction of propagation is randomized after travelling several scattering mean free paths, e.g. the mean distances they travel before their direction of propagation is significantly altered. This situation corresponds to the conditions in which SEA is applied for *bounded* structures. In this case, the scattering mean free paths are presumably small with respect to the characteristic dimensions of the structure. It should be noted however that in infinitely extended homogeneous media, scattering mean free paths are infinite and then a diffusive behaviour can theoretically never be reached. This is in contradiction with the usual assumptions used by the vibrational conductivity approach as it stands to date. The correct setting for the validity of the heat conduction analogy in slender structures such as shells can be established within the frame of the present theory.

The scattering mean free path is defined for a particular propagating mode α by:

$$\ell_\alpha(\mathbf{x}, \mathbf{k}) = \frac{|\nabla_{\mathbf{k}} \lambda_\alpha|}{|\Sigma_\alpha(\mathbf{x}, \mathbf{k})|}. \quad (14)$$

Derivations of the radiative transfer equations have been carried out so far for high frequencies such that $\Lambda \ll c_\alpha t$

and they remain applicable as long as $\Lambda \ll \ell_\alpha$, where Λ is a typical wavelength. The diffusion limit considers that propagation distances are furthermore much longer than the scattering mean free path: $c_\alpha t \gg \ell_\alpha$. It is characterized by a transport mean free path, denoted by ℓ_α^* for the mode α , which is the distance waves must travel until their direction is completely randomized. In this regime and assuming rotational invariance of the scattering cross-sections (they are independent of the direction of the incident rays), coherence matrices $\mathbf{W}_\alpha$ become independent of the direction of $\mathbf{k}$ and are such that $\mathbf{W}_\alpha(\mathbf{x}, \mathbf{k}, t) \rightarrow \Psi(\mathbf{x}, |\mathbf{k}|, t) \mathbf{I}_{R_\alpha}$ where the scalar function Ψ satisfies a diffusion equation:

$$\partial_t \Psi(\mathbf{x}, |\mathbf{k}|, t) = \nabla_{\mathbf{x}} \cdot [D_\alpha(|\mathbf{k}|) \nabla_{\mathbf{x}} \Psi(\mathbf{x}, |\mathbf{k}|, t)] \quad (15)$$

with the diffusion constant $D_\alpha = \frac{1}{d} c_\alpha \ell_\alpha^*$. The ℓ_α^* 's are determined analytically from the knowledge of the total scattering cross-sections (see [4] for the example of plates).

5. Application to thick shells

The results of the previous sections can be applied to a thick curved shell $\Sigma \times (-\frac{h}{2}, \frac{h}{2})$ starting with the following equations. The (regular) middle surface Σ is parametrized by its curvilinear coordinates (x_1, x_2) in a natural tangent basis $(\mathbf{e}_1, \mathbf{e}_2)$, thus $d = 2$ with the notations of section 2. Tangent motions of the middle surface are denoted by $\mathbf{u}_\Sigma$, w is the normal displacement, and $\boldsymbol{\theta}$ is the vector of changes of slope of the normal to Σ . The normal curvature tensor of the shell is denoted by $\mathcal{B}$. Its inverse trace $(\text{Tr } \mathcal{B})^{-1}$, the mean radius of the shell at each point of Σ , is assumed to be much larger than the thickness h , which in turn is significantly lower than a typical wavelength of the excitations. For a thick circular cylindrical shell of radius R , we have $\mathcal{B} = -\frac{1}{R} \mathbf{e}_2 \otimes \mathbf{e}_2$. Let ϱ be the mass per unit surface of the isotropic material constituting the shell, E be its Young's modulus, G its shear modulus such that $G = \frac{E}{2(1+\nu)}$, $\kappa < 1$ its shear reduction coefficient and ν its Poisson's coefficient. The membrane force, shear force and bending moment are denoted by $\mathbf{Q}$, $\mathbf{T}$ and $\mathbf{M}$ respectively. If we let $\mathbf{u} = (\partial_t \mathbf{u}_\Sigma, \partial_t w, \partial_t \boldsymbol{\theta}, \mathbf{Q}, \mathbf{T}, \mathbf{M})$, then $n = 13$ and shell equations for compressional, shear and bending motions together with the constitutive equations linking $\mathbf{Q}$, $\mathbf{T}$ and $\mathbf{M}$ with the strain tensor (see for instance [8]) are equivalent to the system (1) with:

$$\mathbf{A}(\mathbf{x}, 0) = \text{diag} \left(\varrho \mathbf{I}_3, \frac{\varrho I}{h} \mathbf{I}_2, (CE)^{-1}, \frac{1}{Gh}, \frac{1}{\kappa Gh} \mathbf{I}_2, (DE)^{-1}, \frac{1}{GI} \right) \quad (16)$$

where

$$\mathbf{E} = \begin{bmatrix} 1 & \nu \\ \nu & 1 \end{bmatrix}, \quad C = \frac{Eh}{1-\nu^2}, \quad D = \frac{EI}{1-\nu^2}, \quad I = \frac{h^3}{12}.$$

Its dispersion matrix has eigenvalues on the set $\mathbb{R}^2 \times \mathbb{R}^2 \setminus \{(\mathbf{x}, \mathbf{k}); \mathbf{k} = \mathbf{0}\}$:

$$\begin{aligned} \lambda_0 &= 0 && \text{with multiplicity 3,} \\ \lambda_T^\pm &= \pm c_T |\mathbf{k}| && \text{each with multiplicity 1,} \\ \lambda_S^\pm &= \pm c_S |\mathbf{k}| && \text{each with multiplicity 2,} \\ \lambda_P^\pm &= \pm c_P |\mathbf{k}| && \text{each with multiplicity 2,} \end{aligned} \quad (17)$$

with:

$$c_S^2 = \frac{Gh}{\varrho}, \quad c_T^2 = \kappa c_S^2, \quad c_P^2 = \frac{C}{\varrho}$$

and $|\mathbf{k}| = \sqrt{k_1^2 + k_2^2}$ if $\mathbf{k} = (k_1, k_2)^T$. The transport equations are eq. (9) for $\alpha = 0, T^\pm, S^\pm$ or $P^\pm$ with $\mathbf{N}_0 \equiv [\mathbf{0}]_{3 \times 3}$ and $\mathbf{N}_P^\pm = \mathbf{N}_S^\pm \equiv [\mathbf{0}]_{2 \times 2}$. Expressions of the scattering cross-sections for a random material are similar to the ones given in [4] for a Mindlin plate.

6. References

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